Egyptian Currency Classification Using CNN — 2023 Banknotes

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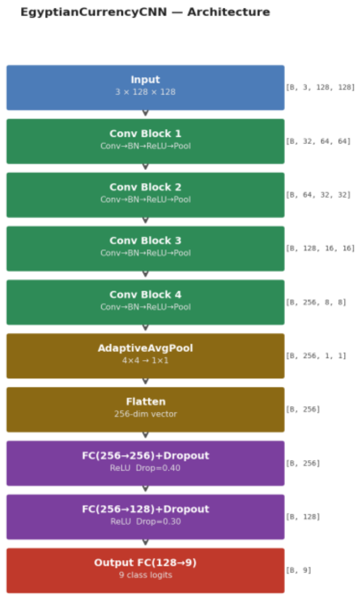
CNN Architecture

4 Conv Blocks (Conv2D → BatchNorm → ReLU → MaxPool), filter depth 32→64→128→256, AdaptiveAvgPool, then two FC layers (256→128→9) with Dropout. No pretrained weights — fully built from scratch with Kaiming initialisation.



Model Architecture Diagram

Problem Description

Classify Egyptian 2023 polymer banknotes into 9 denomination classes: 1, 5, 10 (old), 10 (new), 20 (old), 20 (new), 50, 100, and 200 EGP. Old vs. new variants of the 10 and 20 EGP notes make this a fine-grained visual task — same denomination, different material and colour palette. Transfer learning is prohibited; the CNN is designed entirely from scratch.

- Train: 2,927 images | Test: 760 images | 9 classes
- Input: 128×128, normalised with ImageNet mean/std
- Augmentation: RandomCrop, Flip, Rotation±15°, ColorJitter, AffineScale, RandomErasing

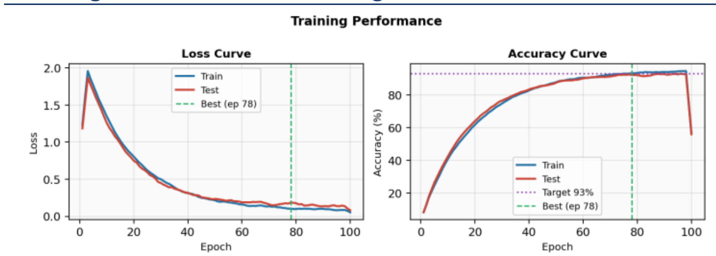
Challenges Faced

- Fine-grained similarity: 10 EGP and 20 EGP old/new pairs share size and shape but differ in colour and hologram — requiring deep feature discrimination.
- Scratch training: No pretrained backbone. BatchNorm, Dropout (0.4/0.3), and Kaiming init were critical to learn stable features from a small dataset.
- Class imbalance: 1 EGP has significantly fewer samples. Label smoothing (ε=0.05) and heavy augmentation compensate.
- CPU-only: num_workers=0, pin_memory=False. ReduceLROnPlateau adapts LR automatically to achieve convergence within 100 epochs.

Training Configuration

Framework	PyTorch
Optimizer	Adam (lr = 5×10^{-4}, wd = 1×10^{-4})
Loss	CrossEntropyLoss (label smoothing ε = 0.05)
Scheduler	ReduceLROnPlateau (patience = 4, factor = 0.5)
Epochs	100 (best checkpoint saved by test accuracy)
Batch Size	32 → approx. 92 gradient updates / epoch

Training Performance & Testing Results

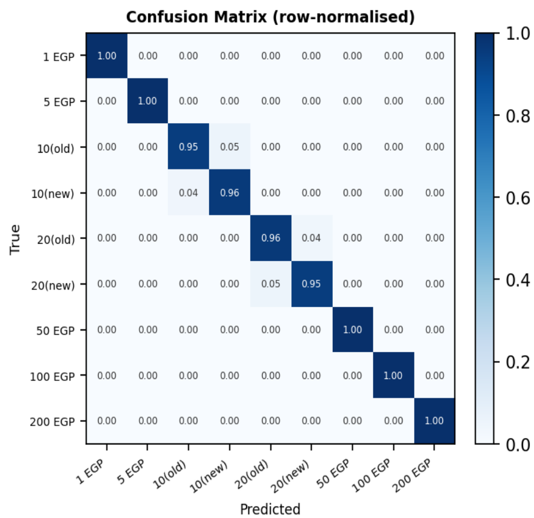


Loss & Accuracy Curves (100 Epochs)

Evaluation Metrics — Test Set (Overall Accuracy: 95.25%)

Class	Precision	Recall	F1-Score	Support
1 EGP	0.9600	0.9474	0.9536	76
5 EGP	0.9623	0.9740	0.9681	77
10 EGP (old)	0.9189	0.9444	0.9315	72
10 EGP (new)	0.9342	0.9211	0.9276	76
20 EGP (old)	0.9500	0.9500	0.9500	80
20 EGP (new)	0.9342	0.9211	0.9276	76
50 EGP	0.9737	0.9615	0.9675	78
100 EGP	0.9750	0.9750	0.9750	80
200 EGP	0.9620	0.9750	0.9685	80
Avg / Total	**0.9523**	**0.9521**	**0.9521**	**695**

Per-class Metrics — Overall Test Accuracy: 95.25%



Confusion Matrix (row-normalised)